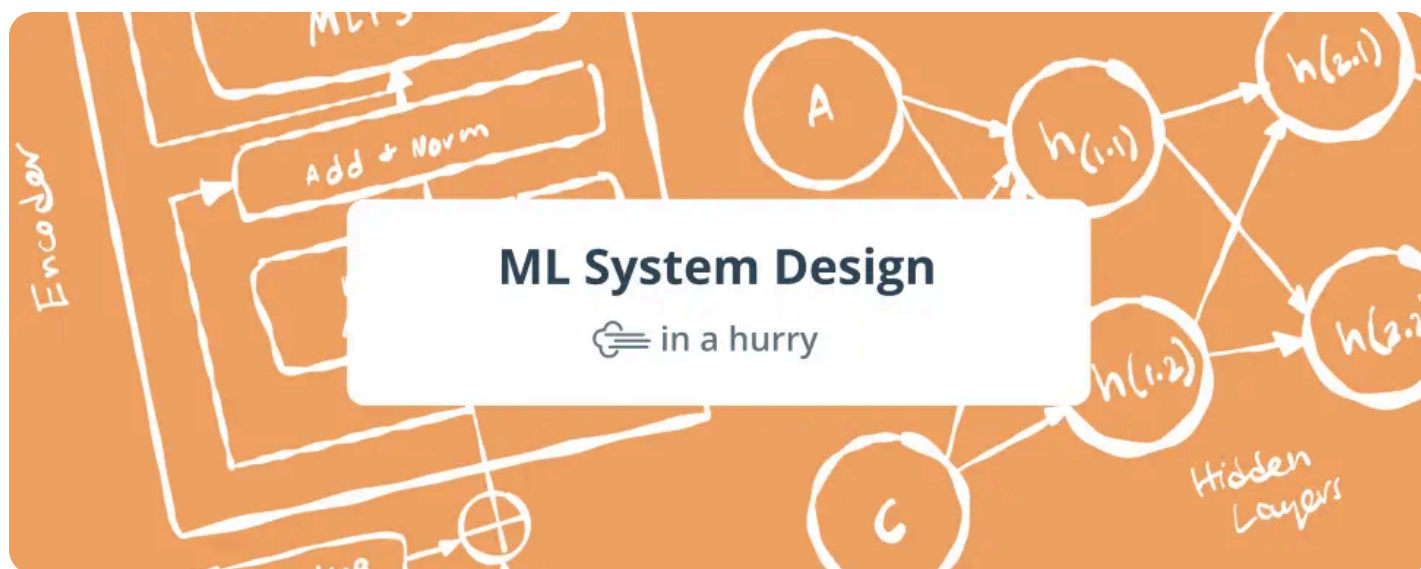




ML System Design in a Hurry

Early Access

Introduction 🗝️

The essentials needed to pass the machine learning (ML) system design interview, built by FAANG hiring managers and staff engineers.

Machine Learning System Design is the wild west of interviews: the field is new and growing rapidly, yet consistency amongst companies and interviewers is low. This makes it frustratingly difficult to be ready for your interview and leads to a lot of anxiety amongst candidates.

Our goal with ML System Design in a Hurry is to give you a framework for navigating these tricky interviews together with the base knowledge you'll need to pass them. This is a practical guide, we're not going to cover the theory of machine learning here. Instead we'll focus on those important skills you'll need to design and implement ML systems in production and pass your interview.

You may find lock icons for some of the content that goes into more depth or teaches you advanced techniques that might be helpful for your interview. Purchasing access to our [Premium](#) content will unlock these for you. But our goal is that free content will be sufficient (with the right amount of additional study) for you to pass your interview.

Ready?

How to Use This Guide

At a high level, the preparation for ML system design interviews about putting pieces together. We'll start with a delivery framework which will help you structure your thoughts and discussion during the interview. On top of this we'll layer in the core concepts, key research, and common patterns you'll need to know.

Once you have the basics, the key will be to practice. We'll walk through some popular problems together to show you how great candidates approach them and what interviewers are looking for.

For your preparation, we recommend that you read this guide in order, skipping any sections you already know. While we link off to additional material where relevant, we've tried to make this guide as self-contained as possible. Don't worry if you don't have time to read the additional material.

Finally, practice with a friend or mentor whenever you can. ML system design interviews are very interactive and how you respond to probes from the interviewer can have just as much weight as the overall direction of your solution.

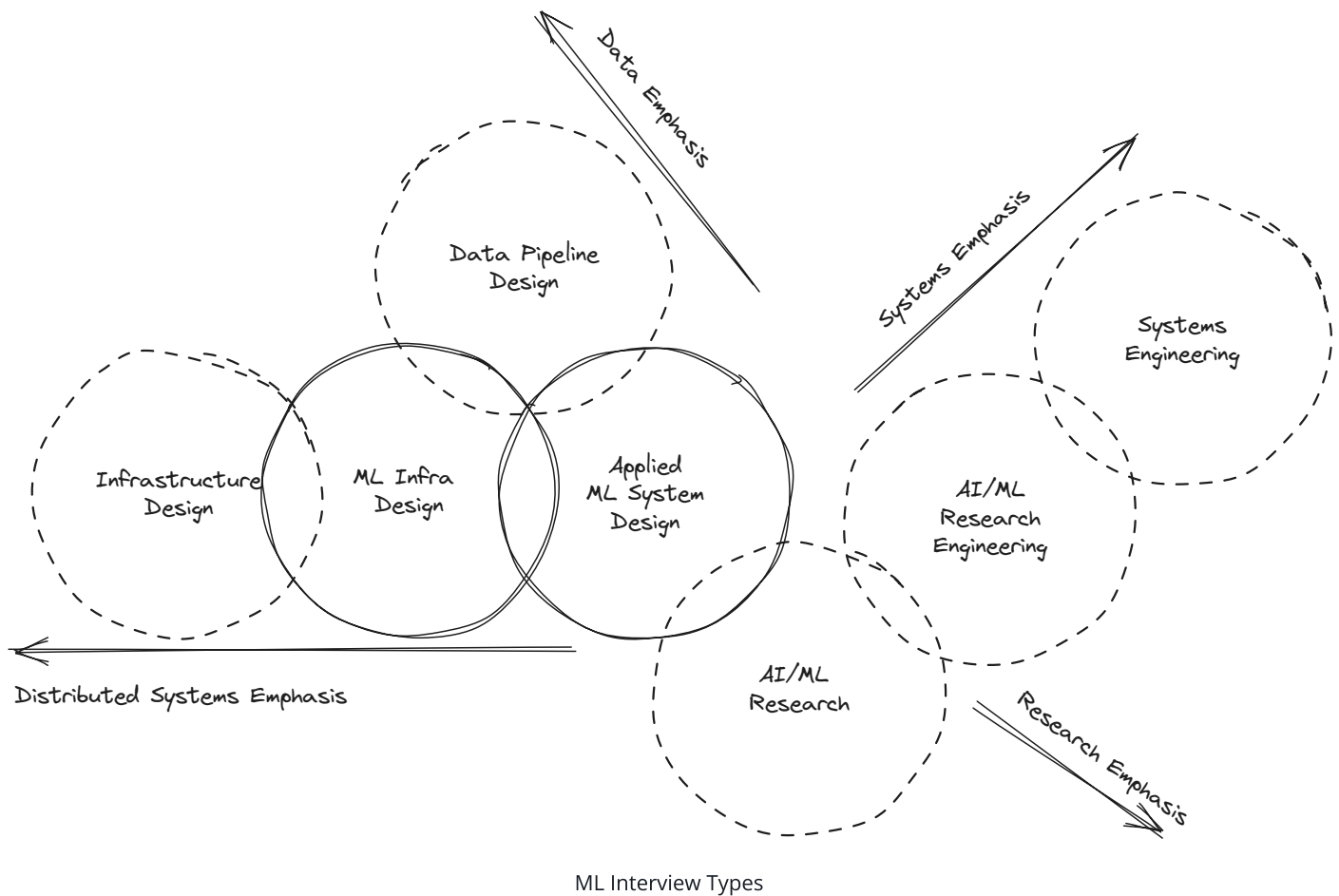


ML System Design in a Hurry is currently in early access. We're iterating rapidly on it and will be adding more content over the coming months. As such, you may find some sections are still under construction. We'll get to them as soon as we can!

Types of ML Interviews

The role of "ML Engineer" is poorly standardized. Companies vary in their expectations of these engineers and, by proxy, their interviews. Spend time with your recruiter to understand what the role is at the company you're interviewing with, you'll be able to work backwards from there to understand what your interview will cover.

We see 4 broad types of ML interviews:



This guide will focus on the most common type of ML interview: Applied ML System Design, but to establish exactly what that means let's talk about the different types of interviews.



The large variance in interview types between companies sometimes leads candidates to trying to pre-qualify themselves in the interview "I've only worked on recommendation systems" or "my team was focused mostly on implementation".

These are good discussions to have with the recruiter **before** your interview, but rarely hold up well during your actual interviews. Remember that the companies are strongly incentivized to avoid wasting their own time on the wrong candidates, so if you got the interview they think there's a chance you're the right candidate.

But prefixing your interview with your uncertainty, however well-intentioned, is likely to undersell yourself before your interviewer can assess your skills. Don't poison the well.

Applied ML System Design

Applied ML System Design interviews focus on your ability to design and implement practical machine learning solutions in production environments, usually presuming access to garden-variety ML infrastructure (e.g. model serving, data pipelines, etc.). These interviews typically assess:

- Your ability to translate business problems into ML solutions
- Understanding of data requirements, preprocessing, and feature engineering
- Knowledge of model selection, training, and evaluation
- Expertise in deployment strategies and monitoring
- Awareness of common pitfalls and how to address them

Common questions might include:

- Design a recommendation system for an e-commerce platform
- Build a fraud detection system for a financial service
- Create a content moderation system for a social media platform

These interviews make up a majority of ML engineering interviews and are the primary focus of this guide.

ML Infra Design

ML Infra Design interviews are focused on the infrastructure of ML systems. These interviews will focus on the challenges of building scalable, performant, and reliable ML systems.

Key areas typically covered include:

- Model serving architectures and scaling strategies
- Training infrastructure design and optimization
- Feature store design and implementation
- ML pipeline orchestration
- Resource management and cost optimization
- Model versioning and deployment strategies

Common questions might include:

- Design a distributed training system
- Create a feature store for a large-scale ML platform

- Design a model serving system that can handle millions of requests per second

These interviews are more common at larger tech companies or ML-focused startups where infrastructure scalability is critical.

This guide is **partly** useful for ML Infra Design interviews, but will have a deeper emphasis on modelling and data design than most ML infra interviews.

AI/ML Research

Research interviews are focused on the theoretical underpinnings of AI/ML. These interviews assess your understanding of:

- Deep learning architectures and their mathematical foundations
- Latest research papers and state-of-the-art approaches
- Ability to design novel algorithms or improve existing ones
- Understanding of optimization techniques and loss functions
- Knowledge of current limitations and research directions

These interviews are most common for research scientist positions at companies with significant R&D investments in AI/ML.

This guide is **not** useful for AI/ML research interviews.

AI/ML Research Engineering

Research Engineering interviews are focused on the intersection of AI/ML and software engineering. Research engineers are often paired with AI/ML scientists to rapidly prototype and test new ideas, they are also frequently responsible for optimizing the performance of ML systems (usually with the objective of proving viability for a research paper).

Key areas of focus include:

- Implementing and optimizing research papers
- Proficiency with ML frameworks and hardware acceleration
- Experimental design and validation
- Performance optimization and benchmarking

- Ability to translate research ideas into working code

These roles are particularly common in AI research labs and companies pushing the boundaries of ML capabilities.

This guide is **not** useful for AI/ML research engineering interviews.

Interview Assessment

Entry-level ML roles are rare (unless you have an advanced degree), and they usually exclude an ML system design component. By mid-level, ML system design interviews become more common, and at the senior level they are the norm.

While each company has its own rubric, ML system design interviews overlap heavily. Interviewers typically evaluate:

- Your ability to turn an ambiguous business problem into an ML solution
- Your practical experience with ML systems—especially spotting the highest-leverage improvements
- Your depth of knowledge in the latest techniques and research

An example rubric might break down the various aspects of the role of an ML engineer, and assess the signal you're giving on each in your interview.

Problem Navigation

First, interviewers want to see whether you can frame a vague business goal as a measurable ML problem. Strong candidates translate product objectives into clear success metrics, state key assumptions and risks, and decide quickly whether ML is even appropriate. Expect to discuss alternative formulations (classification vs. ranking, regression vs. forecasting) and to justify your choice in terms of business impact, data availability, and operational constraints.

More senior candidates will be asked more ambiguous problems, or be expected to navigate and find optimal formulations themselves.

Input Data, Features, and Labels

Next, interviewers are looking to see how effectively you can recruit the right data to solve your problem. This includes designing labels, avoiding leakage, feedback loops, and feature representation discussion (e.g. embeddings, one-hot encoding, etc.).

Seasoned candidates recognize the utility of data and are able to both (a) make use of more data, and (b) have stronger hypotheses about what data is important and how it can be represented in the modelling problem.

Model Design

Model design is the heart of ML system design interviews. Here, interviewers want to see whether you can design a model that is both effective and efficient. This includes selecting the right model, discussions about its architecture, and understanding the trade-offs between different models. Many problems will have multiple components, and you should be able to discuss how they fit together.

Stronger candidates will have more *generalizable* experience that they can apply to the problem. If you, for instance, have experience with recommendation systems but not with fraud detection, you should be able to explain how your knowledge can be useful even in a new domain.

Integration and Evaluation

Finally, interviewers want to see whether you can integrate your model into a production system. This includes discussing how to deploy your model, how to monitor it, and how to iterate on it. Generally speaking there is a gulf between a good idea in a notebook and what works in production and interviewers want to see you can bridge this gap.

A separate discussion about evaluation is a critical part of ML system design and each problem will require a different evaluation strategy. While your interviewer isn't expecting you to be an expert at evaluating all systems, they want to see you realizing common pitfalls and organizing around them.

Communication

All interviews (regardless of type) will include an implicit assessment of your communication skills. Interviewers want to know you'll make a good colleague, that you can collaborate effectively with them on a problem, and that you'll be able to communicate your ideas clearly.

Feedback and Suggestions

We're constantly updating our content based on your feedback. If you have questions, comments, or suggestions please leave them in the comments below. And thanks in advance!